Tree-split cut-out screen, round 2 — run report

Study `research/ic_tree_split_structured_v2_20261011` (aburan28/crypto). **Registration** commit `8a7109c4` (pushed before the run); cairn registration objective `sha256:49b79374…`. **Run** `runs/registered`, 2026-10-11 22:24:56Z – 23:01:18Z UTC, commit `6506a1e2`, harness binary sha256 `28fdd513…cfdb9`. Host: Claude Code cloud container, 4 vCPU, 15 GB, shared with a concurrent census run; untimed study (the metric is a degree; `secs` logged, not measured). Self-test passed.

What ran

`run.sh` ⟶ `examples/tree_split_structured_v2.rs`, 102 cells requested, **102 completed**, 0 failed; 2 `Z` cells censored (|Z| = 111,628 and 132,870 at n = 15, ℓ = 5: degree 5 needs 174,437 monomials > the 90,000 cap; all arms censored, excluded as registered). Arms: `frob2` (Frobenius-stable kernels in no proper subfield), `nbw` (normal-basis windows), `poly` at n = 23, 25, the object `Z = Sym²(T2)` with controls R and ZR, and the `subfield` gate.

Recorded values

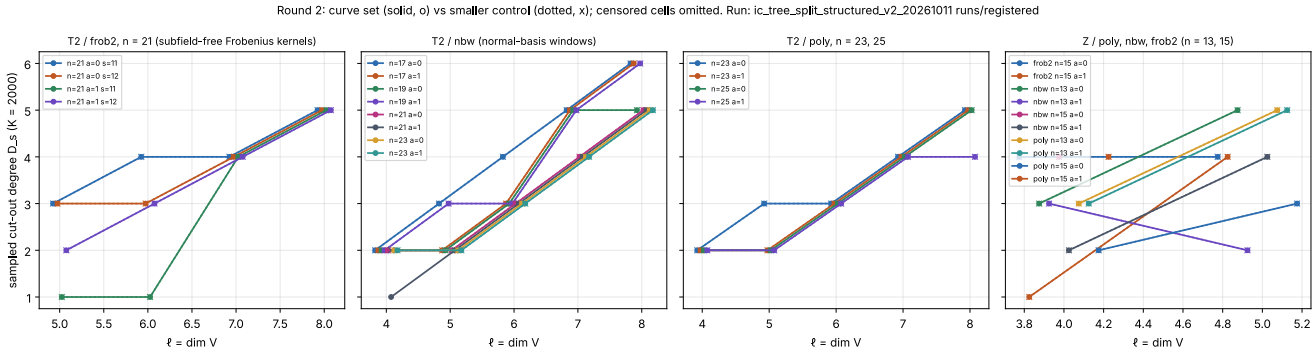


Figure 1: Measured `D_s` against ℓ per series; solid = curve set, dotted = the smaller control. 2,000 samples per arm per degree. Source `runs/registered/cells.jsonl`; plotter `reports/plot_cells.py`.

group	cells (uncensored)	gaps	slopes of D_s(curve) on ℓ	registered verdict
G0 subfield gate	2	T2 ⊆ GF(2^7), deg-1 deficit ≥ 14	—	**PASS**
G1 T2 / frob2, n = 21	16 (16)	all 0	0.6 – 1.5	**CLOSED**
G2 T2 / nbw, n = 17–23	40 (40)	39 × 0, 1 × −1	0.8 – 1.1	**CLOSED**
G3 T2 / poly, n = 23, 25	20 (20)	all 0	0.6 – 0.8	**CLOSED**
G4 Z / poly, n = 13, 15	8 (7)	all 0	no slope clause	**CLOSED**
G5 Z / nbw, n = 13, 15	8 (7)	all 0	no slope clause	**CLOSED**
G6 Z / frob2, n = 15	4 (4)	all 0	no slope clause	**CLOSED**
frob2, n = 35 (supplementary)	4 (4)	all 0	—	not in the verdict

The one −1 (nbw, n = 21, a = 1, ℓ = 4) is a degenerate cell, |F| = |T2| = 4, where the random 4-point control drew one of its 2,000 samples from its 8-point affine closure (probability ≈ 0.4%): control noise.

Checks

Lean `Verdict.lean` (kernel `decide`): the registered rule (`Rule.lean`, sha256 `ec513363…b1e3`) on the 102 recorded cells gives gate = true and CLOSED for G1–G6; mutating a verdict fails. cairn result objective `sha256:84dde9a6…` accepted by the kernel; `cairn audit` (epochs 5 s): chain intact. CI workflow `tree-split-structured-v2-lean.yml` regenerates `Data.lean` and the readout from the records.

Raw records: `runs/registered/{cells.jsonl, cells.log, readout.txt, selftest.txt, commit, sha256}`.